MAT 221 - CALCULUS I
Lesson 5 - Reduction Formulas

DR. M. KOORAPETSE and DR. I. Humbu

August 10, 2026

University of Botswana
Department of Mathematics

Example 1

Find the reduction formula I_n and evaluate I_n at the given value of n.

(a) $I_n = \int x^n e^x \, dx$ and I_3 (b) $I_n = \int \cos^n x \, dx$ and I_5

Solution

(a) Let $u = x^n$ and $dv = e^x \, dx$ then $du = nx^{n-1} \, dx$, $v = e^x$ and

$$I_n = \int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx = x^n e^x - nI_{n-1},$$

$$\therefore \; I_n = x^n e^x - nI_{n-1}, \quad n \geqslant 1.$$

$$\begin{aligned} I_3 = \int x^3 e^x \, dx &= x^3 e^x - 3I_2 = x^3 e^x - 3(x^2 e^x - 2I_1) \\ &= x^3 e^x - 3x^2 e^x + 6I_1 = x^3 e^x - 3x^2 e^x + 6(xe^x - I_0) \\ &= x^3 e^x - 3x^2 e^x + 6xe^x - 6e^x + C \end{aligned}$$

(b) $I_n = \int \cos^n x\,dx = \int \cos^{n-1} x \cos x\,dx$. Now we set $u = \cos^{n-1} x$, $dv = \cos x\,dx$ then $du = (n-1)\cos^{n-2} x(-\sin x)\,dx$, $v = \sin x$ and

$$I_n = \sin x \cos^{n-1} x + (n-1)\int \cos^{n-2} x(1-\cos^2 x)\,dx$$

$$= \sin x \cos^{n-1} x + (n-1)I_{n-2} - (n-1)I_n.$$

$$\therefore\ I_n = \frac{1}{n}\sin x \cos^{n-1} x + \frac{n-1}{n} I_{n-2}, \quad n \geqslant 2.$$

$$I_5 = \int \cos^5 x\,dx = \frac{1}{5}\sin x \cos^4 x + \frac{4}{5} I_3$$

$$= \frac{1}{5}\sin x \cos^4 x + \frac{4}{5}\left[\frac{1}{3}\sin x \cos^2 x + \frac{2}{3} I_1\right]$$

But $I_1 = \int \cos x\,dx = \sin x$

$$\therefore\ I_5 = \frac{1}{5}\sin x \cos^4 x + \frac{4}{15}\sin x \cos^2 x + \frac{8}{15}\sin x + C.$$

Example 2

(a) Given that $I_n = \int_0^{\frac{1}{2}} (1-2x)^n e^x \, dx \;\; n \in \mathbb{N}$. Show clearly that

(i) $I_n = 2nI_{n-1} - 1, \; n \geqslant 1.$

(ii) $I_4 = 384\sqrt{e} - 633.$

(b) Given $I_n = \int_1^e x(\ln x)^n \quad n \in \mathbb{N}$. Show clearly that

(i) $2I_n = e^2 - nI_{n-1}, \; n \geqslant 1.$

(ii) $I_4 = \frac{1}{4}(e^2 - 3).$

(c) Given $I_n = \int_0^1 x^n e^{-\frac{1}{2}x^2} dx \quad n \in \mathbb{N}$. Show clearly that

(i) $I_n = (n-1)I_{n-2} - e^{-\frac{1}{2}}, \; n \geqslant 2.$

(ii) $I_5 = 8 - 13e^{-\frac{1}{2}}.$

(d) Given $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx \quad n \in \mathbb{N}$. Show clearly that

(i) $I_n = \frac{n-1}{n} I_{n-2}, \; n \geqslant 2.$

(ii) $\int_0^{\frac{\pi}{2}} \sin^6 x(1 + \cos^2 x)dx = \frac{45\pi}{256}.$

Solutions

In class